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Androgen deprivation triggers a cytokine signaling switch to induce immune suppression and prostate cancer recurrence

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eLife Assessment

This potentially **valuable** work aimed at a better understanding of the mechanisms of response and resistance to androgen deprivation therapy in prostate cancer using genetically engineered mouse models. A key observation relates to the timing of TNF blockage therapy and the concept of a "TNF switch." The **solid** data were collected using conventional approaches and the conclusions are mostly justified, particularly with the inclusion of more detailed statistics in the revision. The work will be of interest to the prostate cancer research community.

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Abstract

Androgen deprivation therapy (ADT) is an effective but not curative treatment for advanced and recurrent prostate cancer (PC). We investigated the mechanisms controlling the response to androgen-deprivation by surgical castration in genetically-engineered mouse models (GEMMs) of PC, using high frequency ultrasound imaging to rigorously measure tumor volume. Castration initially causes almost all tumors to shrink in volume, but many tumors subsequently recur within 5-10 weeks. Blockade of tumor necrosis factor (TNF) signaling a few days in advance of castration surgery, using a TNFR2 ligand trap, prevents regression in a PTEN-deficient GEMM. Following tumor regression, a stem cell-like population within the tumor increases along with TNF protein levels. Tumor cell lines in culture recapitulate these *in vivo* observations, suggesting the possibility that stem cells are the source of TNF. When TNF signaling blockade is administered immediately prior to castration, tumors regress but recurrence is prevented. This implies that a late wave of TNF secretion within the tumor – which coincides with the expression of NFκB regulated genes – drives recurrence. The

inhibition of signaling downstream of an NFκB-regulated protein – chemokine C-C motif ligand 2 (CCL2) – prevents post-castration tumor recurrence, phenocopying post-castration (late) TNF signaling blockade. CCL2 was originally identified as a macrophage chemoattractant and indeed at late times after castration gene sets related to chemotaxis and migration are up-regulated. Importantly, enhanced CCL2 signaling during the tumor recurrence phase coincides with an increase in pro-tumorigenic macrophages and a decrease in CD8 T cells, suggesting that recurrence is driven at least in part by tumor immunosuppression. In summary, we demonstrate that a therapy-induced switch in TNF signaling induces an immunosuppressive tumor microenvironment and concomitant tumor recurrence.

Introduction

Androgen deprivation therapy (ADT) via surgical castration (1) and currently via medical therapy employing gonadotropin releasing hormone analogues (2), has been an effective treatment for advanced primary and recurrent prostate cancer (PC) for many decades. Within the past several years, the addition of either the microtubule toxin docetaxel or the androgen biosynthesis inhibitor abiraterone to ADT has improved the medical therapy (3). However, these ADT-combination therapies have only marginally extended overall survival and the fundamental issue for patients with advanced primary and recurrent PC remains the same: ADT (or an ADT-combination) rarely cures PC and resistant tumors are eventually lethal. Thus, there is a need to develop additional therapies for PC, preferably by identifying molecular targets that are mechanistically linked to the development of ADT resistance. ADT is effective because the growth of almost every prostate cancer is dependent on a program of gene expression mediated by the androgen receptor and a set of critical co-regulators. The androgen receptor (AR) gene is activated by genetic alteration in only a small fraction of primary PCs (4). Instead, one or two alterations of a small group of other genes (5) is sufficient to re-program the AR cistrome to transcribe a set of genes that promote immortalization and tumorigenesis (6). Additional alterations in a second class of genes driving proliferation and growth – such as the PTEN tumor suppressor or the c-MYC oncogene – completes the process of transformation to prostate cancer (5).

Tissue reconstitution studies (7) have shown that androgen blockade can act indirectly via soluble factors derived from the tumor micro-environment (TME). Consistent with this paradigm, in which proteins are secreted into the TME and act on the cell of origin (autocrine) or on other cells within the TME (paracrine), we have previously investigated the response of prostate cancers to androgen deprivation (8–13), using cell culture and murine models of human prostate cancer. A recurrent theme of this research has been the role of tumor necrosis factor (TNF) as a pleotropic signaling mediator within the PC TME. TNF binds and trimerizes its cognate receptor and signals via a complex network of intercellular protein-protein interactions to promote either an anti-tumor or pro-tumor fate, depending on the cellular context. In this report, we examine the role of TNF and TNF-regulated genes in the response to castration in a genetically engineered mouse model (GEMM) of PC driven by deletion of the PTEN gene (14). Castration induces regression and subsequent recurrence in this GEMM, resembling the onset of castration resistant prostate cancer (CRPC). Both responses are dependent on TNF signaling and can be separately blocked by differentially timing the administration of a TNF ligand trap molecule. At late times post-castration, when the tumor is beginning to recur in a castration resistant form, TNF induces NFκB-mediated gene expression. Moreover, we demonstrate that the NFκB-induced gene CCL2 – which encodes a protein that is chemotactic for immune suppressive cells such as tumor associated macrophages (TAMs) – is necessary for recurrence in the PTEN-loss model as well as in a second GEMM of PC that is driven by MYC over-expression. Thus, a TNF signaling switch induces immunosuppression in the tumor microenvironment to promote castration-resistant tumor recurrence.

Materials and methods

Animals and tumor imaging

All animal studies were approved by the Roswell Park Comprehensive Cancer Center (RPCCC) institutional animal care and use committee. Male Pb-Cre4; PTEN$^{fl/fl}$ mice (14) and Hi-MYC (15) were breed in the RPCCC laboratory animal shared resource. Mice were housed in environmentally-controlled conditions on a 12-hour light/dark cycle with water and food. For both genetically engineered mouse models (GEMMs) we studied, male pups were genotyped and tumor volumes were monitored by three-dimensional high frequency ultrasound (HFUS) imaging, as previously described (9,16). In brief, mice were anesthetized, the abdomen was depilated and B-mode image scans were acquired using a Vevo 2100 micro-ultrasound imaging system (Vevo LAZR; VisualSonics Inc., Toronto). Images were imported into Amira software (Visualization Sciences Group, Burlington, MA) for 3D tumor reconstruction and volume determination. Tumors developed in ~100% of male animals for both GEMMs. The histology, lobular location and usual age to reach 300-500 mm^3 volume are provided for the two models in Supplementary Table S1. Mice with tumor volumes of 300-500 mm^3 were randomly enrolled into experimental groups, and surgically castrated (or sham-operated) via scrotal incision, under isoflurane anesthesia (17). To systematically define the response of the GEMMs to castration, we applied a murine approximation of the human clinical RECIST (response evaluation criteria in solid tumors) criteria (18). Thus, we define regression as a volume decline of 16% or more post-castration and recurrence (of tumors that had met the regression standard) as occurring when we observe volume increases of 11% from the volume nadir, in at least two temporally distinct imaging sessions. Etanercept (sTNFR2-Fc; Amgen, Thousand Oaks, CA, USA) was injected intra-peritoneally (4mg/kg) twice per week for up to 5 weeks following castration, based on our previous studies (8,11). The CCR2 antagonist (CCR2a) BMS-741672 (Torcis Bioscience, Bristol, UK) was injected intraperitoneally (2 mg/kg) three times per week as described (19), beginning one day prior to castration.

Tumor tissue collection and processing

After mice were sacrificed, individual prostate tumors were identified grossly, excised and cut in half. Typically, a portion was flash frozen in liquid nitrogen and stored in a liquid nitrogen freezer while a second portion was fixed in paraformaldehyde at room temperature overnight, washed in PBS, stored at 4°C and embedded in paraffin as required for the preparation of tissue sections. Mouse prostate tumor and normal tissues were dissociated for single cell RNA sequencing using the Tumor Dissociation kits from Miltenyi Biotec (cat. no. 130-096-730). Briefly, tissue was excised, washed with chilled 1x PBS, placed in a petri dish on ice, cut into 2-4 mm^3 pieces, transferred to a gentleMACS C tube (cat. no. 130-096-334) containing DMEM/enzyme mix and then attached to gentleMACS Octo Dissociator (cat. no. 130-096-427) with heating running the '37C_m_TDK_1' (mouse) program. The suspension was filtered through a 40 µm cell strainer (Falcon). The cells were harvested by centrifugation at 400g for 10 min and resuspended in chilled 1X Red Blood Cell Removal Solution (Miltenyi Biotec, cat. no.130-094-183). Debris was then removed using the Debris Removal Solution (Miltenyi Biotec, cat. no. 130-109-398) following the manufacturer's instructions.

Cell culture

The C4-2 cell line, from Jer-Tsong Hsieh, UT Southwestern Medical Center, Dallas, TX, was used to model castration-resistant prostate cancer (20). C4-2 was periodically monitored by polymerase chain reaction (PCR) for mycoplasma contamination (21) and DNA fingerprinting was employed to authenticate this cell lines (22). C4-2 was cultured in RPMI-1640 media supplemented with 10% FCS and 1% penicillin/streptomycin. Enzalutamide was from Selleck Chemicals (Houston, TX).

Enzyme-linked immunosorbent assays (ELISA)

In the case of prostate tumors, tissue powder was homogenized in NP-40 buffer using Sonic Dismembrator (Fisher Scientific, Hampton, NH) at power 7 for 30 sec. The supernatant was collected by centrifugation and stored at −80°C for subsequent analysis. In the case of cell culture, cells were seeded at a density of 0.8-1×10^5/ml as mono-cultures and media stored at 4°C was used for subsequent analysis. TNF and CCL2 were quantitated by ELISA (eBioscience, San Diego, CA), according to the manufacturer's instructions. Protein concentration was determined using bicinchoninic acid reagent (Pierce, ThermoFisher Scientific), according to the manufacturer's instructions.

Fluorescent activating cell sorting

Cells in culture were detached from cell culture plasticware using StemPro accutase (Gibco, Gaithersburg, MD), washed with PBS and incubated with 1% BSA in 1x PBS for 10 min at 4°C. Human cell lines were incubated with antibodies against CD166-FITC (3A6; Ancell, Stillwater, MN; 1:1250) and CD49f-Alexa Fluor® 647 (GoH3; Biolegend, San Diego, CA; 1:200) for 30 min at 4°C. Mouse prostates were disaggregated using 1mg/ml type I collagenase and 1mg/ml Dispase (Invitrogen, Carlsbad, CA) in RPMI medium and filtered through 40 μm cell strainer. Cells were suspended in DMEM with 10% FBS and 1% L-glutamine, and then incubated with fluor-labeled antibodies for 30 min at 4°C. Mouse cells were incubated with antibodies against Sca-1-APC (D7; 1:500), CD49f-PE (GoH3; 1:333), and the 'lineage' markers (Ter119-FITC (TER-119; 1:250), CD31-FITC (390; 1:250) and CD45-FITC (30-F11; 1:250)). LSC FACS, first gating on lineage-negative (Lin^-) cells and then CD49f and SCA-1 was performed as described by Witte and colleagues (23). All anti-mouse antibodies were purchased from eBioscience (San Diego, CA). 7-AAD (Invitrogen, Carlsbad, CA; 1:1000) was used to determine dead cells. FACS analysis was performed using the BD LSR-II (BD Bioscience, San Jose, CA) and analyzed by Winlist 3D (Version 8.0; Verity Software House, Topsham, ME, USA). Cell sorting was done by BD FACSAria-II (BD Bioscience, San Jose, CA).

Colony formation assay

Cells were seeded in 6-well plates at densities of 0.5×10^3, 1×10^3 and 1.5×10^3 cells per well and cultured for 21 days. The media was removed, wells were washed once with PBS, stained with 2% toluidine blue for 30 min and scanned with Epson Perfection 1240U (Suwa, Nagano Prefecture, Japan). Colonies were imaged and then enumerated using ImageJ.

Immunohistochemistry

Immunohistochemical (IHC) staining was performed on 5 μm sections of formalin fixed paraffin-embedded (FFPE) mouse prostate tumor tissue. Sections were incubated with monoclonal antibodies against F4/80 (BM8; Biolegend, San Diego, CA, USA; 1:600), CD8alpha (D4W2Z; Cell Signaling Technology, Danvers, MA, USA; 1:400), or CCL2 (2H5; Biolegend, San Diego, CA, USA; 1:100), appropriate horse-radish peroxide secondary antibodies and reacted with 3,3-diaminobenzidine using an automated tissue processor. IHC and H&E images were captured at 20x magnification using Olympus BX45 microscope and cellSens software (Olympus, Shinjuku-ku, Tokyo, Japan). Ten random high powered (200X) fields were captured for each section and were blinded prior to quantitation of stained cells ($F4/80^+$ or $CCL2^+$ macrophages or $CD8a^+$ lymphocytes) using ImageJ.

RNA extraction, sequencing and analysis

RNA was extracted from liquid nitrogen frozen tumor samples by pulverizing tissue into a powder using a pre-cooled mortar and pestle. TRIzol (Invitrogen, Carlsbad, CA) was then added by a ratio of 1 mL of TRIzol per 100 mg of tumor tissue, and RNA extraction completed as per manufacturer's instructions (24). Bulk RNA-seq data processing and analysis was performed as follows. FastQC v0.11.5 was used for quality control; STAR v2.6.0a (25) was employed to align the reads to the murine reference genome (GRCm38 M16) and gene expression was then quantified using RSEM (26). FastQC, RSEM, STAR were all performed in a local Linux/Unix computing environment. Subsequent analyses were performed using the following R-software packages from the Bioconductor website. Estimated gene counts from RSEM were filtered and the upper quartile was normalized using edgeR (27). Differential gene expression (DGE) was performed using Limma after Voom transformation followed by linear regression (28,29). The significance threshold was set to false discovery rate (FDR) / adjusted p-value of less than 0.05 with a fold change greater than 2 (FDR < 0.05 and |log2 fold change| > 1). Sets of genes with FDR<0.05 and |log2FC| from the DGE analysis were subjected to Gene Ontology analysis of biological processes as described (30), using the website tool (*http://pantherdb.org/index.jsp*). Heatmaps were generated using the package Complexheatmap (31). Single cell gene expression libraries were generated using the 10X Genomics Chromium platform and sequenced on a NovaSeq 6000 (Illumina). Processing of scRNA-seq data was performed as follows. For QC, normalization, clustering and differential gene expression, the Cellranger v6.0.0 software (*https://support.10xgenomics.com/single-cell-geneexpression/software/*) was used to align FASTQ sequencing reads to the mm10 (mouse) reference transcriptome, generating single cell feature counts for each sample. Count data from all cells were included in a 'SingleCellExperiment' (version 1.8.0) object (32). The 'scater' package (version 1.14.5) was used to filter cells and normalize counts per million (CPM) (33). Transcripts from a given cell were removed from the analysis if the corresponding transcriptome met any of the following criteria: 20% or more reads aligned to mitochondrial genes, 1000 total counts per cell or less, or 300 genes per cell or less. The dimensionality of this transcriptomic data was reduced using the t-distributed stochastic neighbor embedding (t-SNE) algorithm (34,35). The 'limma' package (29) was used to identify differentially expressed genes to refine the assignment of clusters generated by the t-SNE algorithm.

Statistics

Data are presented as the mean ± SEM. Statistical analyses were conducted using JMP-Pro12 (SAS, Cary, NC, USA). Differences between two means was assessed by Student's unpaired *t*-test and differences among multiple means by one-way or two-way ANOVA followed by Tukey-Kramer HSD post-hoc testing if the F-statistic was significant. For the small *n*, categorical data sets related to the studies using GEM prostate cancer models shown in **Figures 4A**, 5C and 5D, we employed the exact binomial test to determine the Clopper-Pearson confidence interval for the proportion and Fisher's exact test to determine the *p*-values. Supplementary Table S3 summarizes these statistical analyses.

Results

Tumor recurrence following castration-induced regression in PTEN-deficient PC

Mice engineered to delete the PTEN gene in the prostate epithelium at puberty uniformly develop an intraductal form of prostate adenocarcinoma (14) (Table S1). To investigate the response of prostate cancer to androgen deprivation therapies, we used high frequency ultrasound (HFUS) and three-dimension reconstruction software (9,16) to serially determine the volume of tumors in live animals (**Fig. 1A**). As seen in **Figure 1B**, the kinetics of tumor volume following castration is complex. Immediately following castration, tumor volume continues to increase similar to the non-castrated control. We recently demonstrated that this delay in tumor regression – which is not observed in the normal murine prostate – is driven by glucocorticoid receptor signaling transiently substituting for tumor-specific re-programmed AR signaling (9). Following this delay, the tumors regressed as expected but within ~5 weeks most began to increase in volume, implying tumor recurrence and mimicking the development of CRPC in men. Regression and recurrence events were defined by adapting standard human oncology imaging protocols to mice, as described in the MATERIALS AND METHODS section and in the legend to Supplementary Table S2. Serial HFUS imaging of a large cohort castrated PTEN-deficient mice (including those in **Fig. 1**) is summarized in Supplementary Table S2, demonstrating 40-80% recurrence dependent on the period of post-castration observation.

We measured TNF protein levels in tumor lysates, and found an increase at 5 weeks post-castration, when tumor recurrence is beginning (**Fig. 1C**). Previously, we had observed an increase in *Tnf* mRNA at 8 hours post-castration in the stromal compartment of the normal prostate, suggesting a relatively direct effect of androgen-withdrawal on TNF expression (11). In the case of PTEN-deficient tumors, the increase in TNF protein at a late time post-castration suggests a more complex transcriptional and/or translational regulatory mechanism in response to androgen withdrawal. Alternately, the increase in TNF at late times post-castration might be due to the expansion of a population of TNF-expressing cells as the tumor evolves in response to androgen deprivation. We queried gene expression profiles in public databases from two previously published studies of men who had either hormone-naïve PC or CRPC (36,37), to gain insight into this question. Increased *TNF* mRNA expression in CRPC, relative to castration-naive primary PC, correlated with elevated expression of stem cell-related genes in both datasets (Fig. S1).

Androgen-deprivation induced TNF secretion is due to increased basal cell stemness

To determine if we could detect a change in the stem cell-like tumor epithelial cell populations as the tumors began to increase in volume following castration-induced regression, we examined single cell preparations from tumors for cell surface expression of stem cell markers, using fluorescent-activated cell sorting (FACS). We observed an increase in $Lin^-/Sca1^+/CD49f^{hi}$ (LSC^{hi}) cells at 25 days post-castration, relative to non-castrated tumors (**Fig 1D**). These cells correspond to the cytokeratin-5^+/$p63^+$ basal stem cell population (38), a multipotent stem cell population that is required for the post-natal development of the prostate (39). Basal stem cells are distinct from $Lin^-/Sca1^+/CD49f^{med}$ (LSC^{med}) luminal progenitor cells (identified in the lower right population in **Fig. 1D**) that are the likely tumor initiating cell populations for most human prostate cancers (39,40).

The data in **Figures 1** and S1 suggest that the TNF secreted during the post-castration recurrence phase in the PTEN-deficient tumors might be derived from a cell population with increased levels of 'stemness'. To test this more thoroughly, we performed experiments using the C4-2 and LNCaP cell culture models of PC. First, we observed that long-term culture in the anti-androgen enzalutamide (41) significantly reduced the growth rate of C4-2 prostate cancer cells (**Fig. 2A**), as expected. Reduced growth was accompanied by an increase in the fraction of cells that resembled basal stem cells (**Fig. 2B-C**) and a coordinate increased in TNF levels (**Fig. 2C**). In the analyses for **Figure 2**, we employed CD166 as a basal stemness biomarker (42). We also performed the enzalutamide selection experiment on the LNCaP cell line, an isogenic parental line for C4-2 (20), which is more androgen-responsive (Fig S2A). We observed a similar correlation between stemness and TNF secretion (Fig. S2B-C). Finally, we replicated enzalutamide selection for both C4-2 and LNCaP cell lines, using immunofluorescence microscopy to identify $CD49f^{hi}$ and $CD166^{hi}$ cells (Fig. S3). Again, an increase in the fraction of cells scoring for these stemness markers correlated with increased TNF secretion. Stem cell marker enrichment >10% was typically required to detect an increase in TNF in the media of enzalutamide-treated cultures. Next, we treated C4-2 cells with enzalutamide for 20 days and used preparative FACS to partially purified cells into $CD166^{lo}$ and $CD166^{hi}$ fractions (**Fig. 3A**). TNF levels were higher in conditioned media from the $CD166^{hi}$ fraction relative to the $CD166^{lo}$ fraction (**Fig. 3B**). We also demonstrated that the efficiency of colony formation – a functional marker of stemness – correlated with the level of expression of CD166 (**Fig. 3C**).

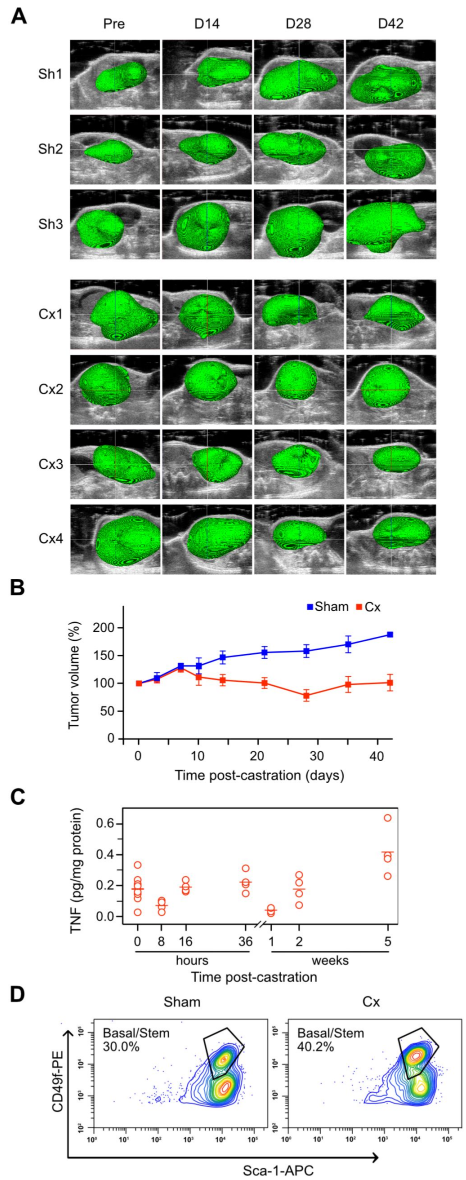


Figure 1.

Castration induces tumor recurrence, and an increase in TNF and stemness in prostate tumors from PTEN-deficient mice.

Mice bearing PTEN-deficient tumors were either castrated (Cx1-4) or sham-operated (Sh1-3) and serially imaged using HFUS. In some cases, tumor tissue was recovered for further analysis. **A**. 3D reconstruction of tumor-bearing ventral lobes (pseudo-colored green) performed as described in Materials & Methods. **B**. Tumor volume kinetics post-castration. Tumor volume was determined from HFUS images using Amira software. Mean tumor volume at the indicated times post-castration (n=4, red) or post-sham operation (n=3, blue) is shown as percent pre-castration volume (mean [%] ± SEM). **C**. TNF levels in tumors tissue extracts were measured by ELISA and normalized to total protein in the tissue sample, at the indicated times post castration. One-way ANOVA followed by Tukey-Kramer HSD test revealed a difference in TNF levels between 0 and 5 weeks (p<0.01). **D**. LSC FACS analysis of tumor cells from sham (left) and castrated (Cx; right) PTEN-deficient mice. Tumors were disaggregated and cells were processed for Lin^-/Sca-1/CD49f FACS analysis (23). Relative fluorescence intensities are shown as contour plots. The LSC^{hi} population is labeled 'Basal/stem'. The population in the lower right quadrant (LSC^{med}) corresponds to luminal progenitors. A representative analysis is shown. Sham LSC FACS was replicated 5 times (mean[%]±SEM = 27.0±4.0); 25d post-castration LSC FACS was replicated 4 times (mean[%]±SEM = 38.2±1.7). Student's t-test revealed a difference between Sham and Cx in the LSC^{hi} population (p<0.05).

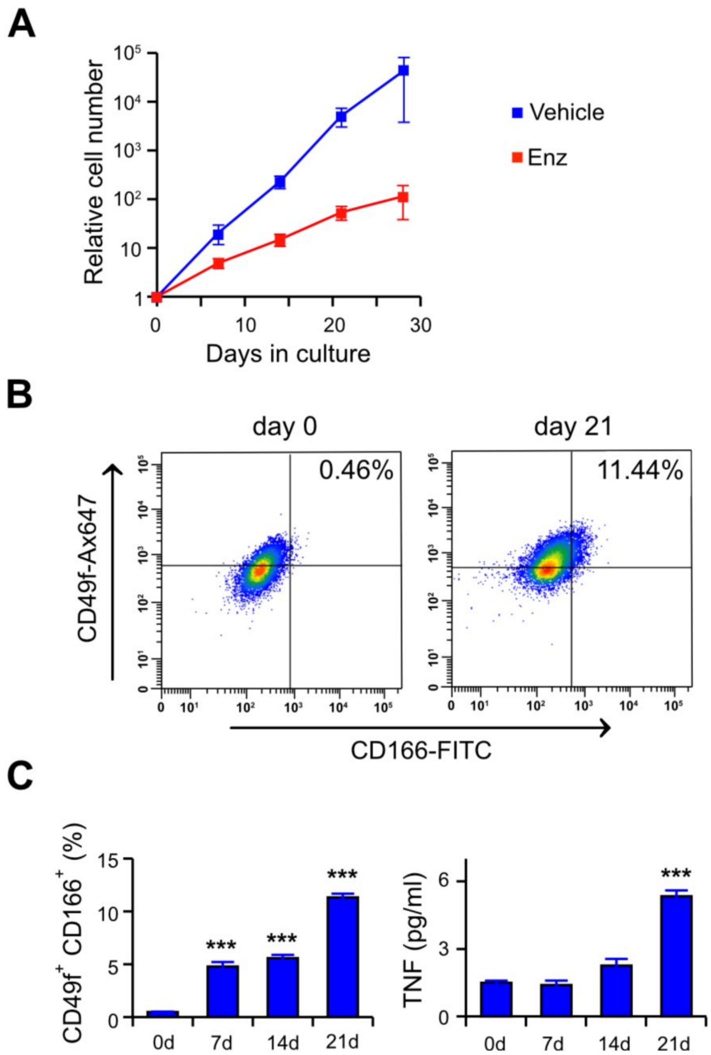


Figure 2.

Extended enzalutamide treatment selects for basal stemness and TNF secretion.

C4-2 cells were grown in media plus 10% serum and treated with vehicle (blue) or 10 μM enzalutamide (Enz; red) for the indicated time. **A.** C4-2 cell growth curve in the presence of enzalutamide. Trypan-blue excluding live cells were counted microscopically at the indicated times. Data are shown as mean ± s.e.m. (n=3). Two-way ANOVA followed by Tukey-Kramer HSD test revealed a difference in cell number for the Enz treated culture at 28d versus 0d (p<0.01). **B.** FACS analysis for basal cell stemness markers in enzalutamide treated C4-2 cells. Cells grown in enzalutamide for the indicated time, were incubated with the fluor-labeled antibodies, analyzed by FACS and relative fluorescence intensity represented as dot plots. The fraction of the cells that correspond to the CD49f^{hi}/CD166^{hi} population (upper, right quadrant) is indicated. Additional time points are in Supplementary Figure S2. **C.** Fraction of cells scoring CD49f^{hi}/CD166^{hi} and corresponding TNF secretion, at the indicated time of culture in media plus enzalutamide. Data from FACS analyses (see Fig. S2) is plotted on the left and TNF levels in the media determined by ELISA is plotted on the right. Values are given as mean ± s.e.m. (n = 2). **, p < 0.01 ***p < 0.001 (two-way ANOVA followed by Tukey-Kramer HSD test).

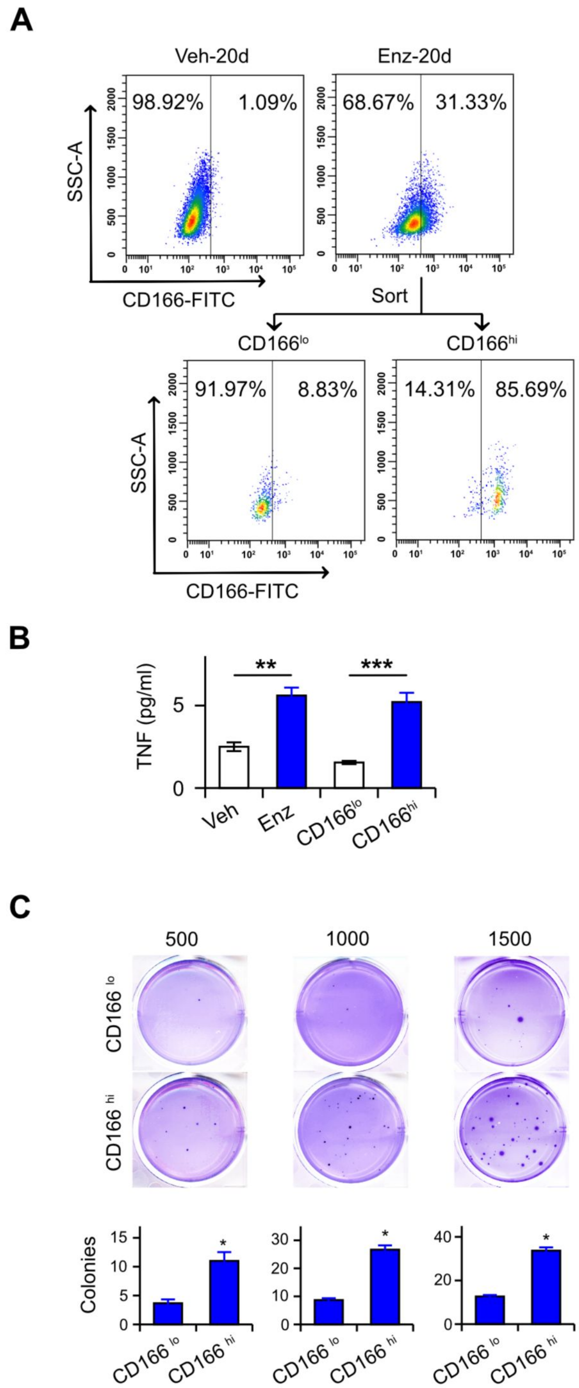


Figure 3.

Enzalutamide treatment selects for a TNF-expressing basal stem cell population.

C4-2 cells were grown for 20 days in media plus 10% serum containing 10 μM enzalutamide, enriched for CD166^{hi} expression. ELISA was used to measure TNF and colony formation. **A.** CD166-enrichment was performed via FACS to sort CD166^{hi} and CD166^{lo} C4-2 cells treated as indicated. The purity of the unsorted and sorted CD166^{hi} and CD166^{lo} populations are shown. **B.** TNF secretion of the indicated cultures (left to right): unsorted, vehicle-treated (veh); unsorted, enzalutamide-treated (enz); and sorted, enzalutamide-treated (CD166^{lo}, CD166^{hi}). TNF measured by ELISA, and plotted as mean ± s.e.m. (n=3). **p<0.01, ***p<0.001 (One-way ANOVA followed by Tukey-Kramer HSD test). **C.** Colony formation assay of enriched CD166^{hi} and CD166^{lo} cell populations sorted as in **A**. Seeding densities were 500, 1000, and 1500 cells/well, respectively. Colony counts per well are plotted. *p<0.05 (Student's unpaired *t*-test).

Differentially-timed blockade of TNF signaling

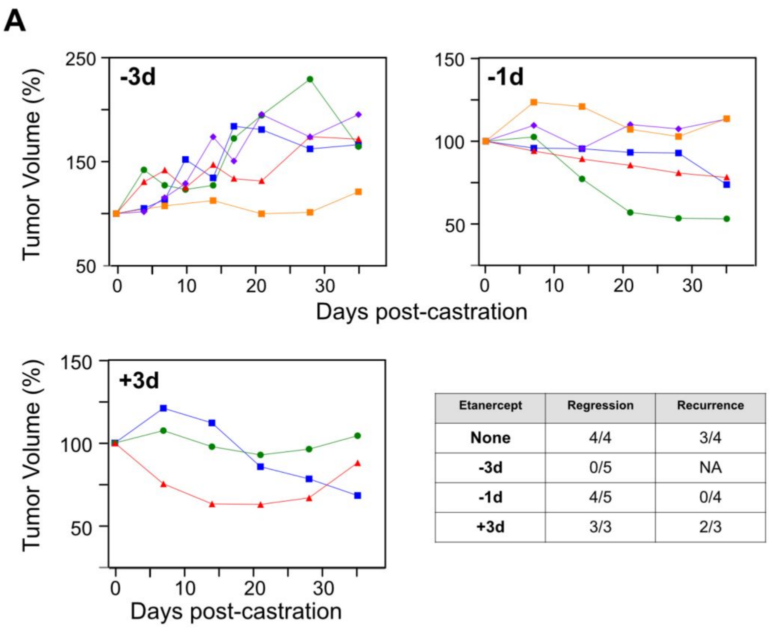
We previously employed cell culture and mouse models of prostate cancer to demonstrate that TNF signaling mediates two distinct responses to ADT – an anti-tumorigenic apoptotic/regression phenotype (8,43) and a pro-tumorigenic cell migration phenotype (44). Given the observations in **Figure 1**, we hypothesized that TNF mediates both the regression and recurrence phases that occur after castration (**Fig. 1B**). To test this, we treated tumor-bearing mice with etanercept – a sTNFR2-Fc fusion protein, which competitively binds TNF (45) and prevents receptor binding – beginning at three distinct time points: three days prior, one day prior, and three days following castration. When etanercept is administered three days before castration (**Fig. 4A**, -3d), castration-induced regression is blocked, and tumor growth is similar to the sham group (**Fig. 1B**). Etanercept given one day before castration does not affect regression but blocks recurrence (**Fig. 4A**, -1d). Finally, when etanercept was administered three days after castration there is no detectable effect on regression or recurrence (**Fig. 4A**, +3d). Instead, the result resembles castration alone (**Fig. 1B**).

To gain insight into the nature of the different molecular processes occurring during regression and recurrence, we performed gene set enrichment analysis (GSEA) on RNA sequence data from bulk tumor mRNA. Specifically, we compared samples from sham castrated mice with those from either mice six days after castration (**Fig. 4B**, upper panel) or mice 35 days after castration (**Fig. 4B**, lower panel). Multiple gene sets related to TNF-induced transcription, most NFkB-related, were uniformly down-regulated at six days post-castration but up-regulated 5 weeks after castration, relative to the sham conditions. The switch in TNF-related gene sets from down-regulation to up-regulation is consistent with a switch from apoptosis signaling (low NFkB levels) to signaling that promotes transcription of NFkB-regulated genes (high NFkB levels).

CCL2 signaling is required for recurrence in murine prostate cancer models

CCL2 is a chemotactic cytokine (chemokine) (46) linked to prostate cancer pathogenesis (47). The CCL2-CCR2 signaling axis has multiple pro-tumorigenic functions, most notably the recruitment and polarization of tumor associated macrophages (TAMs) (48). Previously we demonstrated autocrine CCL2 secretion following enzalutamide induction of TNF in mono-cultures of C4-2 cells (10), consistent with CCL2 gene transcription via TNF induction of NFκB (49). We returned to the enzalutamide selection experiments shown in **Figure 2C** and Supplementary Figure S2 and found that the secretion of CCL2 protein increased over time, in a manner dependent on secreted TNF, in enzalutamide treated C4-2 and LNCaP cells (Fig. S4A-B)To assess paracrine CCL2 production in response to androgen deprivation, we replicated these experiments using co-cultures of LNCaP or C4-2 plus the myofibroblast-like WPMY-1 cell line (50), grown in enzalutamide-containing media. Again, we observed coordinate secretion of TNF and CCL2 (Fig. S4C). Co-cultures produced 5- to 20-fold more CCL2 than the prostate cancer cell line mono-cultures. We also found that enzalutamide-treated co-cultures of C4-2 plus WPMY-1 produced conditioned media that induced migration of THP-1, a CCR2$^+$ macrophage-like leukemia cell line that is a model for TAMs (Fig. S4D).

These observations suggested that paracrine TNF induction of CCL2 in the prostate cancer TME might be responsible for tumor recurrence at late times post-castration, possibly by recruiting immunosuppressive TAMs (51). Indeed, we found that in the samples from **Figure 1C**, CCL2 increased at 5 weeks (**Fig. 5A**), coincident with the rise in TNF that accompanies tumor recurrence. Tissue sections corresponding to tumors 5 weeks post-castration showed an increase in stroma-localized macrophage-like cells expressing CCL2 (**Fig. 5B**). Moreover, CCL2 expression was reduced in tumor tissue sections from castrated mice that received etanercept, consistent with



Etanercept	Regression	Recurrence
None	4/4	3/4
-3d	0/5	NA
-1d	4/5	0/4
+3d	3/3	2/3

B

Sham vs CX_6d

Genesets	SIZE	NES	p.val	FDR
PHONG_TNF_RESPONSE_VIA_P38_PARTIAL	144	-2.98	<0.01	5.16E-05
PHONG_TNF_RESPONSE_VIA_P38_COMPLETE	208	-2.94	<0.01	5.65E-05
RUAN_RESPONSE_TO_TNF_TROGLITAZONE_DN	39	-2.70	<0.01	2.97E-04
PHONG_TNF_TARGETS_UP	60	-2.65	<0.01	3.72E-04
RUAN_RESPONSE_TO_TNF_DN	80	-2.62	<0.01	4.81E-04
ZHOU_TNF_SIGNALING_30MIN	51	-1.97	<0.01	2.35E-02
ZHANG_RESPONSE_TO_IKK_INHIBITOR_AND_TNF_UP	190	-1.88	0.01	3.52E-02
ZHOU_TNF_SIGNALING_4HR	49	-1.85	0.01	4.09E-02
WANG_TNF_TARGETS	18	-1.77	0.02	5.82E-02
RUAN_RESPONSE_TO_TNF_TROGLITAZONE_UP	15	-1.68	0.03	8.09E-02

Sham vs CX_35d

Genesets	SIZE	NES	p.val	FDR
ZHANG_RESPONSE_TO_IKK_INHIBITOR_AND_TNF_UP	193	3.96	<0.01	9.35E-06
PHONG_TNF_TARGETS_UP	61	3.96	<0.01	9.35E-06
PHONG_TNF_RESPONSE_NOT_VIA_P38	307	3.72	<0.01	9.35E-06
WANG_TNF_TARGETS	19	3.5032	<0.01	9.35E-06
TIAN_TNF_SIGNALING_VIA_NFKB	27	3.2884	<0.01	9.35E-06
TIAN_TNF_SIGNALING_NOT_VIA_NFKB	21	2.7396	<0.01	1.64E-04
SANA_TNF_SIGNALING_UP	65	2.7088	<0.01	2.21E-04
PHONG_TNF_RESPONSE_VIA_P38_PARTIAL	149	2.3381	<0.01	2.27E-03
SANA_TNF_SIGNALING_DN	82	2.1602	<0.01	6.18E-03
ZHOU_TNF_SIGNALING_30MIN	51	1.8286	<0.01	3.32E-02
ZHOU_TNF_SIGNALING_4HR	49	1.7752	0.02	4.19E-02

Figure 4.

Differentially-timed TNF signaling blockade has distinct effects on tumor growth.

Mice with PTEN-deficient tumors were castrated and etanercept (sTNFR2-Fc) was administered at the indicated times: three days before (-3d), one day before (-1d), or three days after (+3d) surgical castration. **A.** Tumor volume kinetics. Tumor volumes were determined by serial HFUS imaging and 3D reconstruction and plotted after normalizing to the initial tumor volume for each mouse. Different colors/symbols represent individual mice. The table summarizes regression and recurrence incidence for each experiment. Growth curves for castrated mice that did not receive etanercept (None) are not shown. The criteria for regression and recurrence are described in the Materials and Methods. Statistical analysis is provided in Supplementary Table S3A. **B.** Gene set enrichment analysis. RNA was extracted from PTEN-deficient tumors from the following mice: sham-operated, 6d post-castration and 35d post-castration. GSEA was performed on bulk RNA-seq data using gene sets corresponding to TNF-related pathways. Normalized enrichment scores (NES) from RNA from tumors at 6d (down-regulated, blue) and 35d (up-regulated, red) post-castration were compared to the sham group. Gene sets common to the two analyses are highlighted in yellow.

CCL2 regulation by TNF (**Fig. 5B**). To determine if CCL2 signaling was necessary for post-castration tumor recurrence in prostate cancer mouse models, mice bearing PTEN-deficient prostate cancers received an antagonist of CCR2 prior to castration. Serial imaging revealed that none of the tumors met the criteria for recurrence (**Fig. 5C**), indicating that CCL2-CCR2 signaling is indeed required for castration resistant recurrence in this model. Importantly, the tumor kinetics produced by CCR2 blockade phenocopies the effect of etanercept administered one-day prior to castration (**Fig. 4A**, -1d), consistent with TNF regulation of CCL2 gene transcription via NFkB. To confirm the role of CCL2 signaling we replicated the CCR2 inhibition studies in the Hi-MYC prostate cancer GEMM, which has a similar response to castration (regression delay, regression then recurrence) as the PTEN-deficient GEMM (**Fig. 5D**, blue symbols; see also ref. (9)). We observed that CCR2a treatment prior to castration was comparably effective at preventing post-castration recurrence in Hi-MYC mice (**Fig. 5D**, red symbols; also see Fig. S5).

We used a transcriptomic approach to further investigate CCL2 signaling, immune cell migration, and TAMs in the development of castration-resistant prostate cancer in the PTEN-deficient GEMM. Differential gene expression (DGE) analysis was performed on bulk RNA-sequencing data from tumor tissue and genes with FDR<0.05 and |log2FC| (see Fig. S6 and details in the legend) were subjected to Gene Ontology (GO) enrichment analysis (**Fig. 6**). These analyses revealed that a very similar category of gene sets – mediating immune cell chemotaxis and migration – are up-regulated in the bulk tumor mRNA from mice 35 days post-castration (**Fig. 6A**) but are down-regulated in the bulk tumor mRNA from mice 35 days post-castration which were treated with CCR2a (**Fig. 6B**). We also assembled a TAM-related gene set and found that these genes are similarly regulated: 'up' in the 35 days post-castration group and 'down' in the 35 days post-castrated group treated with CCR2a (**Fig. 6C**).

An immunosuppressive cell profile is induced in the TME during recurrence

Taken together, the analyses in **Figures 5** and **6** indicate that CCL2 might be functioning to recruit TAMs and drive the progression to, or recurrence as, castration resistant prostate cancer (52). Thus, to determine how TNF-CCL2 signaling effects the tumor immune microenvironment in the PTEN-deficient murine prostate cancer model, we used immunohistochemistry (IHC) to quantitate F4/80- and CD8-stained cells in tissue sections from the tumors. As seen in **Figure 7A**, F4/80-stained cells (presumptive TAMs) increase, most notably at 35 days post-castration, just as tumor re-growth in the recurrence phase is beginning and CCL2 levels are peaking (**Fig. 5A**). In addition, etanercept partially blocks this increase in TAMs and coordinately inhibits tumor recurrence. CD8 T cells (presumptive cytolytic T-lymphocytes; CTLs) decreased over time following castration and this was also partially reversed by etanercept administration to the mice bearing PTEN-deficient tumors. **Figure 7B** demonstrates that blocking CCL2 signaling with the CCR2 antagonist has the expected effect – it reduces F4/80-stained cells and increased CD8-stained cells, reversing the castration-induced immunosuppressive cell profile and coordinately preventing prostate cancer recurrence (**Fig. 5C**). Integrating the data on F4/80-staining, CD8-staining and CCL2 ELISAs for tumors treated and harvested at various times post-castration reveals a demonstrates a high degree of correlation of the key immune components we examined. Specifically, there is positive correlation of macrophages and CCL2 secretion and a negative correlation of T cells with macrophage as well as with CCL2 section (Fig. S7). We also observed a more pronounced CD45+ cell infiltrate in the recurrent tumors, relative to non-recurrent tumors (Fig. S7B).

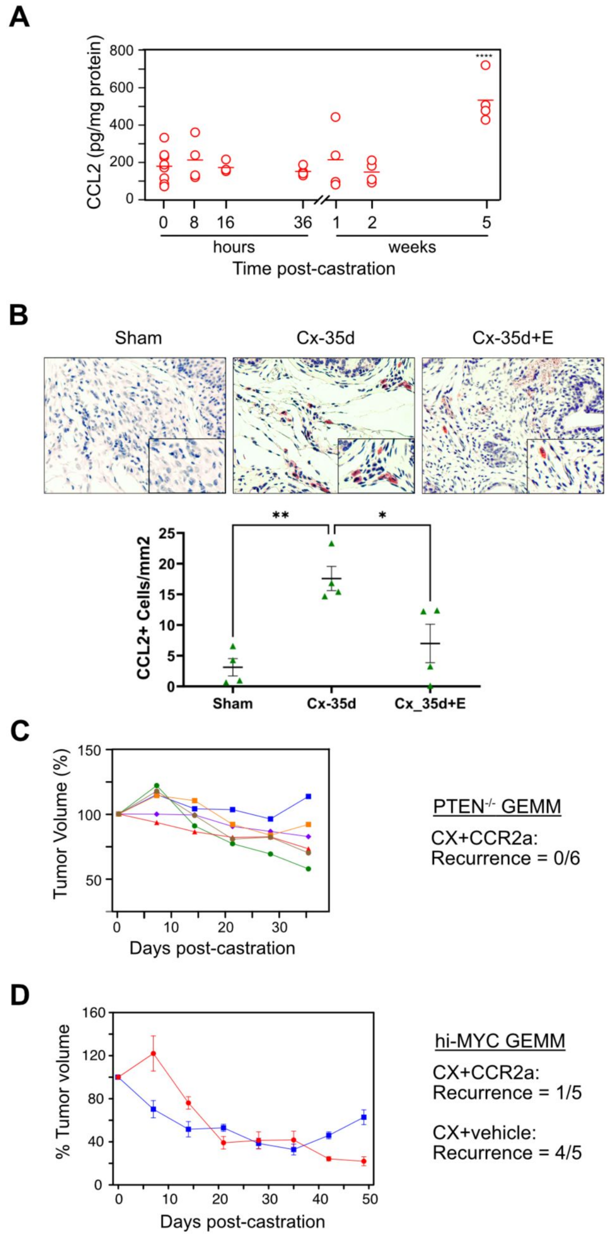


Figure 5.

CCL2-CCR2 signaling is required for castration-resistant recurrence.

Mice bearing PTEN-deficient (**A-C**) or hi-MYC (**D**) tumors were sham-operated or castrated at the indicated times. Volume was monitored by HFUS. Some mice received etanercept (E) or the CCR2a, as indicated. **A**. Tumor CCL2 levels following castration. CCL2 levels were measured by ELISA on the same tumor extracts prepared for **Fig. 1C**, normalized to total protein and plotted as open circles. The bar marks the mean. Two-way ANOVA followed by Tukey-Kramer HSD test revealed a difference in CCL2 levels between 0 and 5 weeks *($p < 0.0001$)*.(**B.** Representative images of IHC-stained CCL2$^+$ cells (red) in tumors from the indicated treatment groups. CCL2+ cells were counted and plotted. **$p < 0.01$, *$p < 0.05$ (Two-way ANOVA followed by Tukey-Kramer HSD test). **C.** Treatment with a CCR2 antagonist phenocopies -1d etanercept treatment (**Fig. 4A**) in PTEN-deficient tumor-bearing mice. Tumor volumes were determined by serial HFUS imaging and 3D reconstruction and plotted after normalizing to the initial tumor volume for each mouse (left). Different colors/symbols represent individual mice. The recurrence incidence is summarized (right). **D.** CCR2 antagonist suppresses castration-induced regression in Hi-MYC prostate cancer-bearing mice. Blue boxes: mice were castrated and received vehicle. Red circles: mice were castrated and received CCR2a prior to castration. N=5 for each group. The recurrence incidence is summarized (right). Tumor volume kinetics for individual mice are in Supplementary Figure S5. Statistical analysis for **C-D** is provided in Supplementary Table S3B.

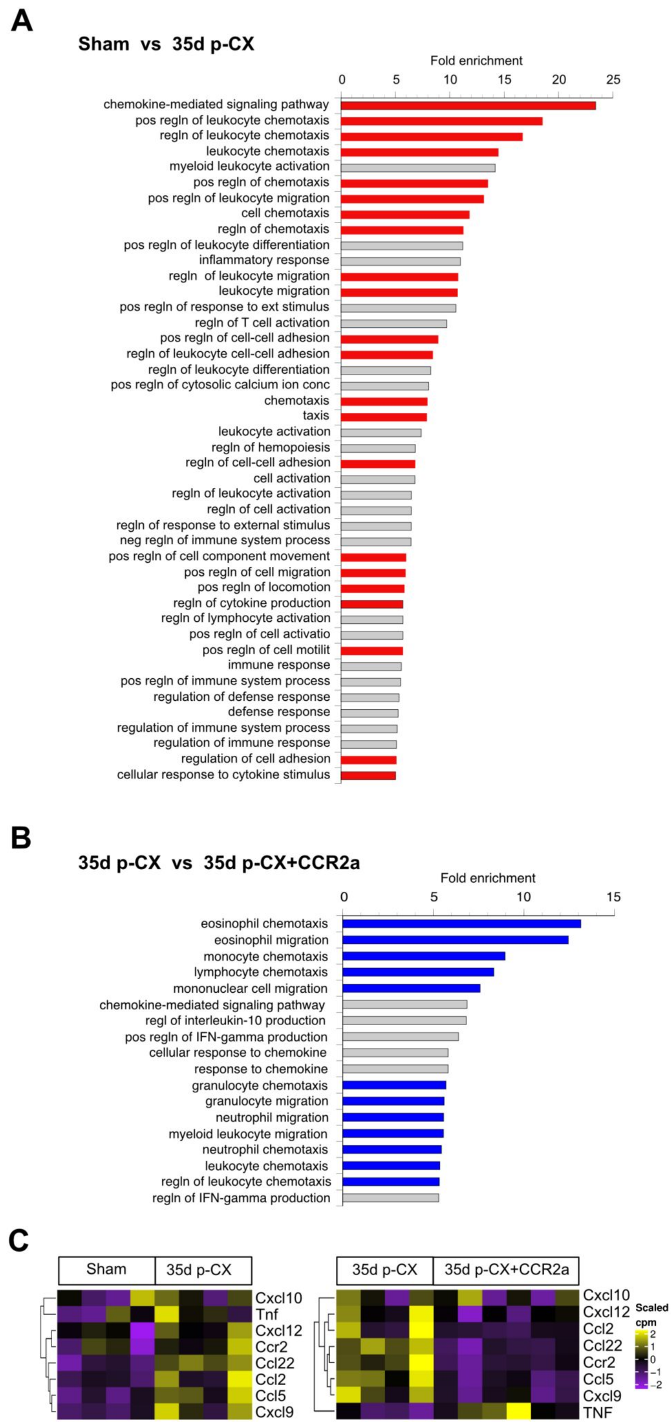


Figure 6.

Transcriptomic changes in PTEN-deficient prostate cancers following CCL2 signaling blockade.

RNA was extracted from PTEN-deficient tumors from the following mice: sham-operated (Sham), 35d post-castration (35d post-CX) and 35d post-castration after receiving CCR2a (35dp-CX+CCR2a). DGE analysis was performed on bulk RNAseq data (Fig. S6) followed by Gene Ontology analysis of biological processes comparing RNAseq transcript sets derived from tumors in mice that were sham-operated vs mice 35d post-castration (**A**) or from tumors in mice 35d post-castration vs mice treated with CCR2a 35d post-castration (**B**). Red-colored bars represent up-regulation and blue-colored bars represent down-regulation. Fold-enrichment >5. **C.** TAM-related transcripts regulated by CCL2 signaling. Heatmap representation of mRNA levels of select TAM-associated genes for the same two comparisons shown in **A** (left) and **B** (right).

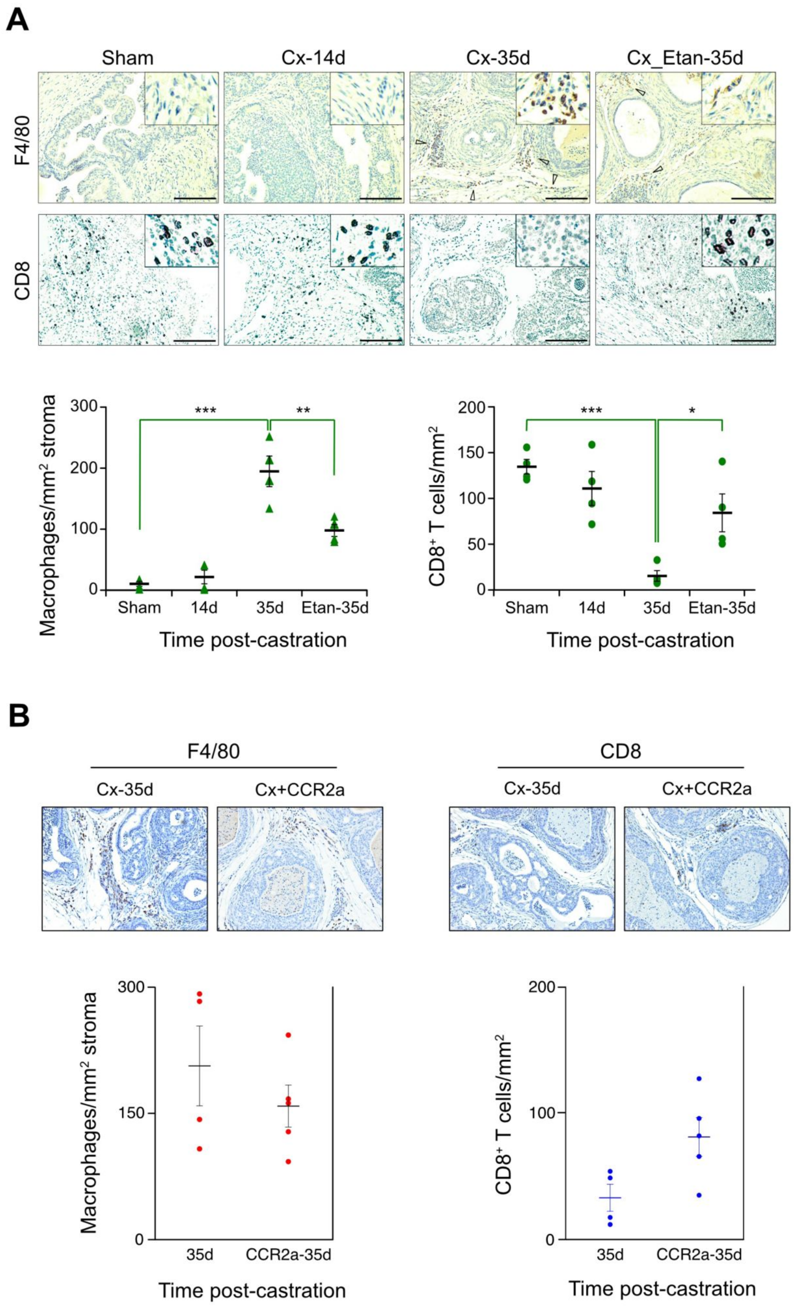


Figure 7.

TNF and CCR2 signaling blockade suppress castration-induced immunosuppression at late times following castration.

Mice bearing PTEN-deficient tumors were sham-operator or castrated (Cx) and tumors harvested at the indicated times post-castration. Some mice received etanercept (Etan) or the CCR antagonist BMS IHC staining for stromal F4/80+ macrophages and CD8+ T cells was performed and the number of stained cells per unit area were plotted. **A**. TNF signaling blockade. Images of representative IHC-stained tissue sections for the surface marker protein noted on the left side. Mice were treated as shown across top and the cell densities plotted for each of the treatments, below. **B**. CCL2 signaling blockade. Similar to **A** with both surface marker protein and corresponding mouse treatments shown along the top. A complete set of images are provided in Supplementary Figure S7C. Staining and counting of the Cx-35d samples in **A** and **B** were performed separately. Data are given as mean ± s.e.m. (n=4). *p<0.05, **p<0.01, ***p<0.001 (One-way ANOVA followed by Tukey-Kramer HSD test). *P*-value for the differences for CD8+ T cells and F4/80 macrophages due to CCR2a treatment was >0.05 (not statistically significant).

Transcriptomic CD8/CD68 ratio predicts castration outcome

Next, we carried out transcriptomic profiling to confirm the IHC studies in **Figure 7**, which correlated an immunosuppressive immune cell profile (elevated CD68, reduced CD8) with recurrence of the PTEN-deficient tumor. Specifically, three mice bearing PTEN-deficient prostate cancers were castrated and tumor volume was HFUS-monitored for 42 days (**Fig. 8A**). One mouse failed to regress and was labeled 'stable'. Two mice regressed, one of which showed recurrence 21 days after castration. We recovered the tumors, and performed single cell RNA sequencing. Transcript count data was pooled and tSNE dimensionality reduction was used, along with subjective cell annotation, to identify six cell populations (**Fig. 8B**). Among CD45-positive immune cells, TAMs were identified as those cells expressing both CD68 and CD80, while cytolytic T cells (CTL) were defined as those expressing CD3 and CD8 (see details in the legend to **Fig. 8**). In **Figure 8C**, we plotted the ratio of CTL to TAM, and observed that compared with the regressing tumor, the recurrent tumor had a lower CTL/TAM ratio. This is consistent with the IHC analysis in **Figure 7A** and suggests that immune suppression drives tumor re-growth.

Discussion

TNF signaling switch

Employing a HFUS-based imaging protocol to measure tumor volume in live mice, we are able to resolve the response of genetically-engineered murine prostate cancers to castration into multiple, mechanistically-distinct phases, as illustrated for the PTEN-deficient prostate cancer model in **Figure 1** (the kinetics for the castration response in the Hi-MYC GEMM are similar). Immediately following castration, there is an increase in tumor volume in both prostate cancer GEMMs. We recently demonstrated that this seemingly paradoxical effect is dependent on glucocorticoid signaling transiently substituting for tumor-specific androgen receptor signals that are blocked by the androgen ablative effects of castration (9). After 4-7 days of gradually increasing volume, the tumors begin to regress due to apoptotic cell death of the epithelial tumor cells in the cancerous glands, consistent with the classical morphological description of apoptosis (53). In addition to tumor epithelial cell apoptosis, there is a loss of vascular integrity within the tumor that contributes to the regression phase (54,55). Some tumors continue to regress for more than two months, but for the majority – up to 75% of individual tumors in the prostate cancer GEMMs we studied – recurrence begins 4-6 weeks after castration (**Fig. 1A**, **Fig 4D** and Supplementary Table S2b).

We sought to decipher the intra- and inter-cellular signaling events that occur within the tumor and the surrounding microenvironment in response to androgen deprivation (castration). Previously we had shown that TNF – but not other common death receptor ligands (FasL and TRAIL) – is necessary for apoptosis-driven regression of the normal murine prostate (11). In addition, we also found that TNF is required for so-called vascular regression of murine prostate cancers, probably by triggering endothelial cell apoptosis (8). The loss of vascular integrity leads to tumor hypoxia within a day following castration. While these events occur multiple days prior to the onset of a measurable reduction in prostate cancer volume, hypoxia may indirectly drive regression by inducing tumor epithelial cell apoptosis via TP53 (8). In the current studies, when we administered etanercept to PTEN-deficient mice three days prior to castration, regression was effectively inhibited (**Fig. 4A** and Supplementary Table S2a). This observation is consistent with TNF-driven apoptosis of tumor epithelial and endothelial cells.

The increase in TNF protein 5 weeks post-castration (**Fig. 1C**) – at a point when most tumors have regressed – suggested a second role for TNF signaling in the response to androgen deprivation. Indeed, blockade of TNF signaling beginning one day prior to castration results in

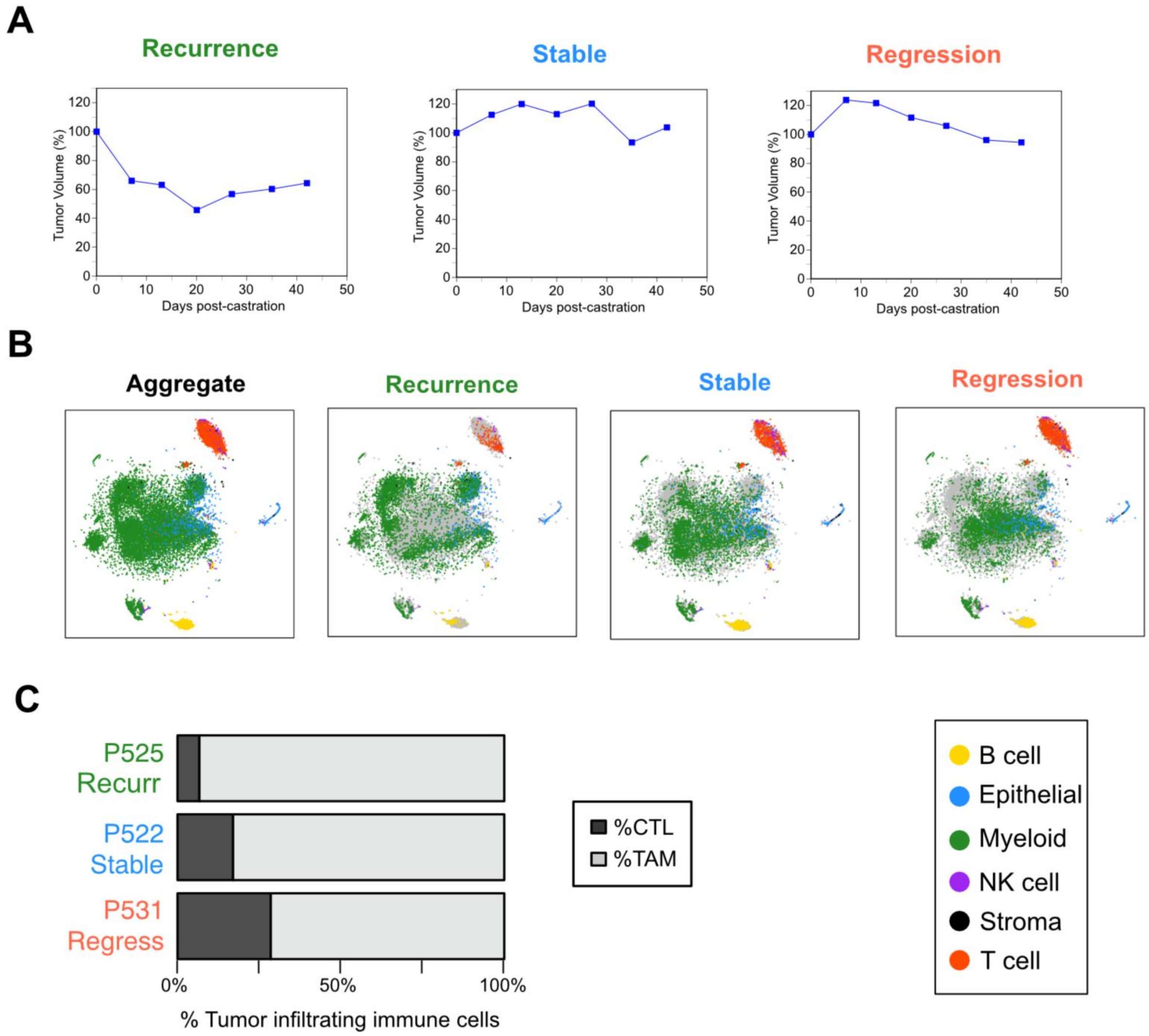


Figure 8.

Tumor recurrence following castration correlates with an immunosuppressive tumor microenvironment.

A. Post-castration tumor volume kinetics for three representative PTEN-deficient mouse tumors. Tumor volumes were determined by serial HFUS imaging and 3D reconstruction and plotted after normalizing to the initial tumor volume for each mouse. The average starting volume for these tumors was ~960 mm^3. **B.** t-SNE plots for single cell RNAseq analysis. At 42d post-castration, mice were sacrificed, tumor tissue was disaggregated into single cell preparations, RNA sequenced and analyzed as described in the Methods. Dimensionality reduction of cell transcriptomes from all three mice was performed with tSNE to produce the plot in far left panel (Aggregate). Cells are assigned to 6 subset populations as coded in the key in the lower right corner. Color-coded cells from each of the three mice corresponding to the response patterns in **A** are shown in the remaining three panels of **B**. Cells which were in the aggregate population, but not identified in individual mice are grey colored. TAMs (defined in panel C) are within the large, heterogeneous myeloid population (green). **C.** CTL:TAM ratios for regressing, stable or recurrent mouse tumors. Transcriptomic data from **B** was used to define TAMs (CD45 and CD68and CD80; lighter grey) and CTLs (CD45 and [CD3e or Cd3d or CD3g] and [CD8a or CD8b1]; darker grey). See text for discussion.

most tumors regressing, but no recurrence (**Fig. 4A** and Supplementary Table S2a). Thus, TNF signaling switches from pro-apoptotic signaling that activates caspase-8 and drives regression to pro-tumorigenic signaling that activates the NFκB and/or PI3K/Akt pathways, promoting tumor cell survival and proliferation and tumor recurrence. The results of transcriptional analysis (**Fig. 4B**) support the switch to a pro-tumorigenic phenotype: we observed increased expression of NFκB-related gene sets in bulk mRNA recovered from tumor samples at 35 days, versus 6 days, post-castration. A key consequence of NFκB signaling during the recurrence phase is the induction of CCL2 protein levels 35 days post-castration (**Fig. 5A**). Moreover, the administration of a CCR2 antagonist to tumor bearing mice prevented castration-induced recurrence (**Fig. 5C**), phenocopying the effects of -1d TNF blockade by etanercept treatment (**Fig. 4A**, -1d panel). This observation is consistent with CCL2 expression downstream of TNF-NFκB signaling. Similarly, the number of CCL2-staining cells in the TME was reduced in tumors from mice receiving etanercept to block TNF signaling (**Fig. 5B**).

Increased tumor cell stemness following androgen deprivation drives pro-tumor TNF signaling

While the pleotropic signaling effects of TNF are well-established, an important mechanistic question in the context of this report is whether changes in the tumor and surrounding TME are responsible for the post-castration switch in TNF signaling. Two potential determinants are the spatial and temporal pattern of TNF production/ secretion and the context of the cells that TNF signaling targets. Previously, we showed that in the normal rat prostate TNF mRNA and protein increased as early as 8 hours post-castration, with the most significant increase occurring within the stromal compartment of the prostate (11). However, in PTEN-deficient murine tumors there is no discernable increase in TNF protein in the first 24 hours post-castration (**Fig. 1B**). Alternatively, death receptor driven apoptosis can be triggered by down-regulation of the CASP8 inhibitor CFLAR (56). Indeed, we previously found castration reduced CFLAR expression within 12 hours post-castration in the normal rat prostate (12,13). Moreover, CFLAR protein is up-regulated in CRPC patient samples (57). To investigate the mechanism of the late (5 weeks post-castration) increase in TNF, we blocked AR action with enzalutamide in prostate cancer cell culture models (10). After 3 weeks of culture in enzalutamide, TNF secretion and stemness were increased in both C42 and LNCaP cell lines (**Figs. 2** and Supplementary Figs. S2-S3). Higher levels of TNF secretion occurred in cultures enriched for expression of one of the stem cell-like markers (**Fig. 3**). The increased expression of stem cell-like genes following androgen deprivation may reflect the death of luminal-like cells (low stemness) and selection for, or expansion of, basal-like cells (higher stemness) that have increased levels of NFκB and therefore secrete more TNF protein. We observed an increase in basal stem-like cells at 25 days post-castration in the PTEN-deficient tumor model (**Fig. 1D**), suggesting a similar phenomenon is occurring *in vivo*. It is also possible that the increase in stemness that we observe post-castration is due to lineage plasticity of pre-existing cells, rather than selection and expansion of *bone fide* tumor stem cells (58). Regardless, it appears most likely that the increase in TNF secretion observed 5 weeks post-castration is a consequence of changes in cellular context rather than a promoter/enhancer-driven transcriptional response.

Immune suppression is a key consequence of increased tumor cell stemness

By investigating the switch in TNF signaling, we identified CCL2 secretion as a key TNF-NFκB-induced event required for the onset of tumor recurrence in the PTEN-deficient GEMM (**Fig. 5**). Moreover, CCL2 is also required for recurrence in the genetically distinct Hi-MYC GEMM, suggesting CCL2 may be a common driver for recurrence post-castration, at least in murine models of prostate cancer. Although CCL2 has multiple pro-tumorigenic immune-modulating functions in prostate and other cancers (59), the transcriptomic studies described in **Figure 6** are consistent with TAM recruitment. Specifically, Gene Ontology analysis (**Fig. 6A-B**) indicates that CCL2 induces immune cell chemotaxis and migration at the time of recurrence, which is the behavior expected of TAMs as well as other myeloid immune suppressive cells, such as myeloid-derived suppressor cells (MDSC). The heat map in **Figure 6C** showed up-regulation of TAM-related genes, while IHC analysis in **Figure 7** (and Supplementary Fig. S7) demonstrated a coordinate increase in F4/80+ macrophages and CCL2 at the time of recurrence. The latter two observations support a critical role for TAMs, but do not rule out a role for MDSC. Identifying the precise tumor immune cell populations within the TME that modulate the response of cancers to therapy is the focus of considerable ongoing research (60). TAMs are a heterogenous set of immune cells which are predominantly pro-tumorigenic (61) and have been implicated in human and murine prostate cancers (62,63). Although an oversimplification of the immune response to cancer, a balance between anti-tumor cytolytic T cells and immunosuppressive myeloid cell populations may predict the success of emerging immunotherapies. Indeed, variants of the CTL/TAM ratio – similar to our observations in **Figures 7** and **8** – are potential prognostic biomarkers in patients with melanoma (64), breast cancer (65) and hepatocellular carcinoma (66).

Prospective: a model for the development of CRPC

The models we employed are anatomically similar to non-metastatic CRPC (nmCRPC) in that the tumor is localized to the prostate, whereas in metastatic CRPC (mCRPC) the TME is derived from the metastatic site, such as bone. Most CRPC is discovered clinically as mCRPC and indeed nmCRPC is likely a transient state that progresses to mCRPC (67). Castration of mice bearing PTEN-deficient or Hi-MYC prostate cancers induces regression and then recurrence – a pattern very similar to that seen in the development of human CRPC. Importantly, HFUS imaging provides real-time volume data to ensure that the complex kinetics of tumor volume changes can be used to accurately categorize the response of each tumor to androgen deprivation. Supplementary Figure S8 is a mechanistic model consistent with our findings. It proposes that TNF secretion into the TME first drives regression and subsequently induces an immunosuppressive state that leads to recurrence. The specific signal may vary depending on the TME context. For example, in some PTEN-deficient GEMMs, IL1ß triggers myeloid immune suppression (68). Most of our analyses were performed on bulk tumor tissue – this represents a significant limitation of the studies in this report. Future studies employing multi-modal single cell analyses will be critical to elucidating the molecular interactions between tumor cells and the myofibroblasts, endothelial cells and immune cells that constitute the TME. These studies are underway in our laboratories. Although the CCL2 antagonist was effective at preventing recurrence in our GEMMs, a monoclonal antibody which binds CCL2 failed in a phase 2 patient trial in mCRPC (69). This may have been due to technical issues with the drug (69), or possibly heterogeneity among human CRPCs. Indeed, Giancotti and colleagues identified three subtypes of CRPC (70) and proposed that one of these – mesenchymal stem-like prostate cancer – might use CCL2 signaling to promote myeloid immune suppression. Thus, in the future patients with this CRPC subtype may be candidates for anti-CCL2 therapies.

Data availability

All data generated or analyzed during this study are included in the manuscript and supporting files; source data files such as ultrasound-derived volumetric data and RNAseq data are available from Dr. Nastiuk upon request.

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Additional files

Supplementary Data

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Joint Public Review:

Summary:

Sha K et al aimed at identifying mechanism of response and resistance to castration in the Pten knock out GEM model. They found elevated levels of TNF overexpressed in castrated tumors associated to an expansion of basal-like stem cells during recurrence, which they show occurring in prostate cancer cells in culture upon enzalutamide treatment. Further, the authors carry on timed dependent analysis of the role of TNF in regression and recurrence to show that TNF regulates both processes. Similarly, CCL2, which the authors had proposed as a chemokine secreted upon TNF induction following enzalutamide treatment, is also shown elevated during recurrence and associate it to the remodeling of an immunosuppressive microenvironment through depletion of T cells and recruitment of TAMs.

Strengths:

The paper exploits a well stablished GEM model to interrogate mechanisms of response to standard of care treatment. This of utmost importance since prostate cancer recurrence after ADT or ARSi marks the onset of an incurable disease stage for which limited treatments exist. The work is relevant in the confirmation that recurrent prostate cancer is mostly an immunologically "cold" tumor with an immunosuppressive immune microenvironment.

Comments on revised version:

The Reviewing Editor has reviewed the response letter and revised manuscript and has the following recommendations (all text revisions) prior to the Version of Record.

More information for Panel 4A:

For the most part, the authors have addressed the statistical concerns raised in the initial review through inclusion of p values in the relevant figure legends. One important exception is Fig 4A which includes some of the most impactful data in the paper. The response letter and the new Fig4A legend refers to statistical in Supp Table 3. I could not find this in the package. Because this is such an important panel, I would urge the authors to include the statistics in the main figure. The display should include a fourth panel with castration alone, as requested by at least one reviewer.

I would also urge the authors to place a schema of the experimental design at the top of the figure to clarify the timing of anti-TNF therapy and the fact that it is administered continuously rather than as a single dose (I was confused by this upon first reading). Last, it is hard to reconcile the curves in the day +3 panel with the conclusion that there is no effect (the red curve in particular).

Include a model cartoon of the TNF switch:

A key concept in the report is the concept of a "TNF switch". I recommend the authors include a model cartoon that lays out this out visually in an easily understandable format. The cartoon in Supp Fig 8 touches on this but is more biochemically focused and does not easily convey the "switch" concept.

Add a "study limitations" paragraph at the end of the discussion:

The authors noted that several other concerns expressed by the reviewers were considered beyond the scope of this report. These include the inclusion of additional tumor response endpoints beyond US-guided assessment of tumor volume (e.g., histology, proliferation markers, etc.) and the purely correlative association of macrophage and T cell infiltration with recurrence, in the absence of immune cell depletion experiments. To this point, the subheading "Immune suppression is a key consequence of increased tumor cell stemness" in the Discussion is too strongly worded.

Similarly, there is no experimental proof that CCL2 from stroma (vs from tumor cell) is required for late relapse. Prior to formal publication, I suggest the authors include a "limitations of the study" paragraph at the end of the discussions that delineates several of these points.

Other points:

For concerns that several reviewers raised about basal versus luminal cells and stemness, the authors have modified the text to soften the conclusions and not assign specific lineage identities.

The answer to the question regarding timing of castration (based on tumor size, not age) needs more detail. This is particularly relevant for the Hi-MYC model that is exquisitely castration sensitive and not known to relapse, except perhaps at very late time points (9-12 months). Surely the authors can include some information on the age range of the mice.

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Author response:

The following is the authors' response to the original reviews.

> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *Sha K et al aimed at identifying the mechanism of response and resistance to castration in the Pten knockout GEM model. They found elevated levels of TNF overexpressed in castrated tumors associated with an expansion of basal-like stem cells during recurrence, which they show occurring in prostate cancer cells in culture upon enzalutamide treatment. Further, the authors carry on a timed dependent analysis of the role of TNF in regression and recurrence to show that TNF regulates both processes. Similarly, CCL2, which the authors had proposed as a chemokine secreted upon TNF induction following enzalutamide treatment, is also shown to be elevated during recurrence and associated with the remodeling of an immunosuppressive microenvironment through depletion of T cells and recruitment of TAMs.*
>
> *Strengths:*
>
> *The paper exploits a well-established GEM model to interrogate mechanisms of response to standard-of-care treatment. This is of utmost importance since prostate cancer recurrence after ADT or ARSi marks the onset of an incurable disease stage for which limited treatments exist. The work is relevant in the confirmation that recurrent prostate cancer is mostly an immunologically "cold" tumor with an immunosuppressive immune microenvironment*
>
> *Weaknesses:*
>
> *While the data is consistent and the conclusions are mostly supported and justified, the findings overall are incremental and of limited novelty. The role of TNF and NF-kB signaling in tumor progression and the role of the CCL2-CCR2 in shaping the immunosuppressive microenvironment are well established.*

We contend there is novelty in: the experimental design; our finding of a TNF signaling 'switch' and the role of androgen-deprivation induced immunosuppression.

> *On the other hand, it is unclear why the authors decided to focus on the basal compartment when there is a wealth of literature suggesting that luminal cells are if not exclusively, surely one of the cells of origin of prostate cancer and responsible for recurrence upon antiandrogen treatment. As a result, most of the later shown data has to be taken with caution as it is not known if the same phenomena occur in the luminal compartment.*

While we appreciate the reviewer's interest in the cancer stem cell biology occurring in the tumor in response to androgen deprivation, our focus in this report is identifying mechanisms that account for a switch in TNF signaling. Specifically, our previous studies showed a rapid increase in TNF mRNA following castration (in the normal murine prostate) but in the current report we also observe an increase in TNF at late times post-castration (in a murine prostate cancer model). We propose that the increase in TNF at late times is due to plasticity (increased stemness) in the tumor cell population, rather than - for example - a change in signal-driven TNF mRNA transcription. While a possible mechanism is expansion of a recurrent tumor stem-cell population, a careful investigation is beyond the scope of this report. Therefore, in the revised manuscript, we have altered the text in multiple places to indicate a suggestive, rather than definitive, role for tumor stem cells. Indeed, we did include caveats regarding the role of tumor stem cells in the original discussion (lines 425-429 in the revised manuscript), and this is now made more explicit in the revised manuscript.

> ***Reviewer #2 (Public Review):***
>
> *Summary:*
>
> *In this study, Sha and Zhang et al. reported that androgen deprivation therapy (ADT) induces a switch to a basal-stemness status, driven by the TNF-CCL2-CCR2 axis. Their results also reveal that enhanced CCL2 coincides with increased macrophages and decreased CD8 T cells, suggesting that ADT resistance may be related to the TNF/CCL2/CCR2-dependent immunosuppressive tumor microenvironment (TME). Overall, this is a very interesting study with a significant amount of data.*
>
> *Strengths:*
>
> *The strengths of the study include various clinically relevant models, cutting-edge technology (such as single-cell RNA-seq), translational potential (TNF and CCR2 inhibitors), and novel insights connecting stemness lineage switch to an immunosuppressive TME. Thus, I believe this work would be of significant interest to the field of prostate cancer and journal readership.*
>
> *Weaknesses:*
>
> *(1) One of the key conclusions/findings of this study is the ADT-induced basal-stemness lineage switch driving ADT resistance. However, most of the presented evidence supporting this conclusion only selects a couple of marker genes. What exacerbates this issue is that different basal-stemness markers were often selected with different results. For example, Figure S1A uses CD166/EZH2 as markers, while Figure S1B uses ITGb1/EZH2. In contrast, Figure 1D uses Sca1/CD49, and Figure 2B-C uses CD49/CD166. Since many basal-stemness lineage gene signatures have been previously established, the study should examine various basal-stemness gene signatures rather than a couple of selected markers. Moreover, why were none of the stemness/basal-gene signatures significantly changed in the GO enrichment analysis in Figure 6A/B?*

Mice and human cells express similar but also partially distinct prostate stem cell markers. For example, Sca1 is predominantly used as a stem cell marker in mice but not in human prostate epithelial cells. CD166 and CD49f are expressed in both human and murine prostate epithelium and therefore we used these in both sets of studies. Also see the response to R1-2.

> *(2) A related weakness is the lack of functional results supporting the stemness lineage switch. Although the authors present colony formation assay results, these could be influenced simply by promoted cell proliferation, which is not a convincing indicator of stemness. To support this key conclusion, widely accepted stemness assays, such as the prostasphere formation assay (in vitro) and Extreme Limiting Dilution Analysis (ELDA) xenograft assay (in vivo), should be carried out.*

See the response to R1-2 and R2-1, above.

> *(3) Another significant concern is that this study uses concurrency to demonstrate a causal relationship in many key results, which is entirely different. For example, Figure S4A and S4B only show increased CCL2 and TNF secretion simultaneously, which cannot support that CCL2 is dependent on TNF. Similarly, Figure 5A only shows that CCL2 increased coincidently with a rise in TNF, which cannot support a causal relationship. To support the causal relationship of this conclusion, it is necessary to show that TNF-KO/KD would abolish the increased CCL2 secretion.*

Regarding Fig. S4A and S4B: We previously demonstrated (Sha et al, 2015; reference 10) that CCL2 secretion is dependent on TNF, in the same cell lines. We have added additional data (new Fig. S4B) in this report to confirm this dependency.

Regarding Fig 5: In Fig 5B we demonstrated that the increase in CCL2-staining cells in recurrent tumors from castrated animals (the equivalent of human CRPC in our model) was significantly inhibited in animals receiving etanercept, demonstrating TNF dependency for CCL2 in this context.

While the use of TNF KO cell lines and animals could provide additional insights, the creation of such cell lines and tumor models is arduous. Moreover, we previously demonstrated that administration of anti-TNF drugs such as etanercept are as effective as the KO phenotypes (Davis et al 2011; ref. 11).

> *(4) Some of the selective data presentations are not explained and are difficult to understand. For example, why does CD49 staining in Figure S3A have data for all four time points, while CD166 in Figure S3D only has data for the last time point (day 21)? Similarly, although several TNF_UP gene signatures were highlighted in Figure 4B, several TNF_DN signatures were also enriched in the same table, such as RUAN_RESPONSE_TO_TNF_DN. What is the explanation for these contrasting results?*

Regarding Fig. S3A and S3D: The cell-staining studies in Fig. S3 are confirmatory of the FACS studies in Figs. 2 and 3. We were not able to stain all of the CD166 time-points for technical reasons (difficulty optimizing the automated staining protocol) but we were able to successfully stain key late time-points, so we have included this data in the supplementary figure. There was no attempt to selectively present data; this was just a practical limitation of the time and funds that we could devote to confirmatory studies.

Regarding Fig 4B: The highlighting identifies a common (i.e., identical) group of gene sets in the two GSEA analyses, demonstrating that these very same gene sets are all up-regulated in one instance, and down-regulated in the other. The 'TNF DN' genes were not identical in the two GSEA analyses and so we cannot draw any conclusions about these. Note that we are scoring the TNF-related genes sets with the 10 largest (positive or negative) normalized enrichment scores (NES), and are not relying on DN or UP designations in the gene set name (identifier). In this analysis up- and down-regulation refers to the sign and magnitude of the NES, not the gene set names.

> ***Reviewer #3 (Public Review):***
>
> *Summary:*
>
> *The current manuscript evaluates the role of TNF in promoting AR targeted therapy regression and subsequent resistance through CCL2 and TAMs. The current evidence supports a correlative role for TNF in promoting cancer cell progression following AR inhibition. Weaknesses include a lack of descriptive methodology of the pre-clinical GEM model experiments and it is not well defined which cell types are impacted in this pre-clinical model which will be quite heterogenous with regards to cancer, normal, and microenvironment cells.*
>
> *Strengths:*
>
> *(1) Appropriate use of pre-clinical models and GEM models to address the scientific questions.*
>
> *(2) Novel finding of TNF and interplay of TAMs in promoting cancer cell progression following AR inhibition.*

> *(3) Potential for developing novel therapeutic strategies to overcome resistance to AR blockade.*
>
> *Weaknesses:*
>
> *(1) There is a lack of description regarding the GEM model experiments - the age at which mice experiments are started.*

Table S1 in the supplementary data summarizes the salient characteristics of the GEM models. Note that as described in the M&M, we selected animals for experimental groups based on the tumor volume (determined by HFUS) and not based on the age of the mouse, since there is some variability in the kinetics of tumor growth in genetically identical mice, as shown by our HFUS observations of hundreds of mice harboring the genetic changes (PTEN loss, MYC gain) in the models we have studied most extensively. Although admittedly an imperfect criteria, we reasoned that tumor volume would be the best surrogate criteria for tumor biology.

> *(2) Tumor volume measurements are provided but in this context, there is no discussion on how the mixed cancer and normal epithelial and microenvironment is impacted by AR therapy which could lead to the subtle changes in tumor volume.*

The reviewer's criticism is well-founded - most of our studies involved bulk analysis, which makes it difficult to probe the cellular interactions within the TME. Future studies - beyond the scope of this report - using single cell technical approaches - are needed to investigate these subtle changes. We have added a statement to this effect to the manuscript (lines 464-468).

> *(3) There are no readouts for target inhibition across the therapeutic pre-clinical trials or dosing time courses.*

The reviewer's criticism is well-founded, since we cannot be 100% certain of drug delivery in the TNF and CCL2 blockade experiments. Two points in this regard. First, with the assistance of institutional veterinarian staff, we have had good success in training multiple scientists (PhD student, technicians) to deliver both biological and small molecule drugs i.p. Second, the observation that the drugs did 'work' in most animals in well-defined experimental protocols strongly suggests that the delivery methodology is reliable. If sporadic delivery failures do occur, this would tend to underestimate the magnitude of the 'positive' (i.e., blocking) effects rather than leading to false negatives.

> *(4) The terminology of regression and resistance appears arbitrary. The data seems to demonstrate a persistence of significant disease that progresses, rather than a robust response with minimal residual disease that recurs within the primary tumor.*

We explain our rationale for the criteria defining regression and recurrence in the M&M and in the legend to Table S2. In the revised version of the manuscript, we now explicitly reference these descriptions in the relevant RESULTS section (lines 222-223). Note that we use the term 'recurrence' rather than 'resistance' as the former does not necessarily imply a particular biological mechanism.

> *(5) It is unclear if the increase in basal-like stem cells is from normal basal cells or cancer cells with a basal stem-like property.*

See the response to R1-2 and R2-1.

> *(6) In the Hi-MYC model, MYC expression is regulated by AR inhibition and is profoundly ARi responsive at early time points.*

We agree that this is the likely mechanism of castration-induced regression (so-called 'MYC addiction') but it is unclear what the reviewer's concern is vis-a-vis our manuscript.

> ***Reviewer #4 (Public Review):***
>
> *In this manuscript by Sha et al. the authors test the role of TNFa in modulating tumor regression/recurrence under therapeutic pressure from castration (or enzalutamide) in both in vitro and in vivo models of prostate cancer. Using the PTEN-null genetic mouse model, they compare the effect of a TNFa ligand trap, etanercept, at various points pre- and post-castration. Their most interesting findings from this experiment were that etanercept given 3 days prior to castration prevented tumor regression, which is a common phenotype seen in these models after castration, but etanercept given 1 day prior to castration prevented prostate cancer recurrence after castration. They go on to perform RNA sequencing on tumors isolated from either sham or castrate mice from two time points post-castration to study acute and delayed transcriptional responses to androgen deprivation. They found enrichment of gene sets containing TNF-targets which initially decrease post-castration but are elevated by 35 days, the time at which tumors recur. The authors conduct a similar set of experiments using human prostate cancer cell lines treated with the androgen receptor inhibitor enzalutamide and observe that drug treatment leads to cells with basal stem-like features that express high levels of TNF. They noticed that CCL2 levels correlate with changes in TNF levels raising the possibility that CCL2 might be a critical downstream effector for disease recurrence. To this end, they treated PTEN-null and hi-MYC castrated mice with a CCR2-antagonist (CCR2a) because CCR2 is one receptor of CCL2 and monitors tumor growth dynamics. Interestingly, upon treatment with CCR2a, tumors did not recur according to their measurements. They go on to demonstrate that the tumors pre-treated with CCR2a had reduced levels of putative TAMs and increased CTLs in the context of TNF or CCR2 inhibition providing a cellular context associated with disease regression. Lastly, they perform single-cell RNA sequencing to further characterize the tumor microenvironment post-castration and report that the ratio of CTLs to TAMs is lower in a recurrent tumor.*
>
> *While the concepts behind the study have merit, the data are incomplete and do not fully support the authors' conclusions. The author's definition of recurrence is subjective given that the amount of disease regression after castration is both variable (Figure 8) and relatively limited*

See the response to R3-4, above.

> *particularly in the PTEN loss model. Critical controls are missing. For example, both drug experiments were completed without treating non-castrate plus drug controls*

In these experiments, we are investigating the effect of anti-TNF or anti-CCL2 therapy on the response to the castration. The appropriate controls are castrated mice which received vehicle or no treatment. The response of intact animals (with tumors still increasing in size) is not only irrelevant to the question we are asking, but also impractical, as the tumor size would be too large for mouse viability.

> *which raises the question of how specific these findings are to castration resistance. No validation was performed to ensure that either the TNF ligand trap or the CCR2 agonist was acting on target.*

See the response to R3-3, above.

> *The single-cell sequencing experiments were done without replicates which raises concern about its interpretation.*

The goal in these experiments is to address a relatively narrow question concerning changes in a few key TAM-associated transcripts versus changes in a few CTL-associated transcripts. This is not meant to provide rigorous single cell transcriptomic analysis that is required - for example - to definitely assess the levels of various cell populations. As noted in R3-2 (and in the DISCUSSION , lines 467-468) future single cell analysis is ongoing, but beyond the scope of this manuscript.

> *At a conceptual level, the authors say that a major cause of disease recurrence in the immunosuppressive TME, but provide little functional data that macrophages and T cells are directly responsible for this phenotype.*

The requirement for CCL2-CCR2 signaling for recurrence suggests that TAMs drive recurrence, presumably due to immunosuppression in the TME. However, CCR2 is expressed by other cell types. Therefore, in future studies we will need to examine the response to additional inhibitors and also employ single cell ‘omics to more thoroughly characterize the changes in the cellular components of the tumor immune microenvironment. Functional analysis of T-cell subsets is an even more formidable experimental challenge.

> *Statistical analyses were performed on only select experiments.*

See the response to R1-3, below.

> *In summary, further work is recommended to support the conclusions of this story.*
>
> ***Reviewer #1 (Recommendations For The Authors):***
>
> *I suggest the authors address the following:*
>
> *(1) Throughout the figures, statistical analysis needs to be made clear including n numbers, replicates, and whether or not differences shown are statistically significant. These includes Figure 1c, and d,; Figure 2 A and B, Figure 3A; Figure 4A; Figure 5A, C and D; Figure 7B.*

We thank the reviewer for identifying these issues and we have inserted statistical analyses into the text as follows:

Figure 1C-D: Statistical analysis added to the legend of Fig. 1.

FIgure 2A: Statistical analysis added to the legend of Fig. 2.

Figures 2B: These are representative FACS scatter plots – the corresponding statistical analysis is shown in Fig. 2C (left panel).

Figure 3A: Statistical comparisons are not relevant to this figure – the data is presented to document the cell sorting enrichment process.

Figure 4A and Figure 5C-D: For the small n, categorical data sets related to the studies using GEM prostate cancer models shown in Figures 4A, 5C and 5D, we employed the exact binomial test to determine the Clopper-Pearson confidence interval for the proportion and Fisher’s exact test to determine the p-values and now present these analyses in a new

Supplementary Table 3. We have included this information in the M&M section and edited the Figure legends to direct the reader to the new Supplementary Table.

We would like to emphasize that the reported p-values are exact probabilities from Fisher's exact test. Given the small sample sizes and the discrete nature of the distribution, these values should not be interpreted as if they strictly conform to conventional thresholds such as $p<0.05$. Instead, they represent the exact probability of observing data as extreme as (or more extreme than) what we obtained under the null hypothesis.

Figure 5A: The legend of Fig. 5A was edited to clarify the statistical analysis.

Figure 7B: The differences in CD8+ T cells and F4/80 macrophages due to CCR2a-35d treatment were not statistically different ($p>0.05$) - we have now stated this explicitly in the figure legend.

> *(2) Several experiments either lack appropriate controls or the choice of data presentation is confusing. In Figure 4A vehicle controls should*

We have not observed any effect of IP administration of vehicle in any experiments across multiple published studies employing these GEMMs, and so we conclude that the injection of vehicle is very unlikely to modify the outcome of these experiments.

> *be included in the graphs and for ease of interpretation perhaps average tumor growth should be shown with individual tumor growth can be shown in the supplement. In Figure 5 the vehicle control is missing and in Figure 5D 4 out of 5 CX+vehicle tumors are said to have recurred but the trend line in the graph shows otherwise.*

We thank the reviewer for noting this issue - the color designations were inadvertently reversed in the legend text. This error has been corrected in the revised version of the manuscript.

> *In Figure 8B flow cytometry would actually be more convincing than scRNAseq. If scRNAseq is chosen, a higher quality UMAP or t_SNE plot is needed with a broader color palette.*

We did consider the FACS approach suggested by the reviewer, but decided against it as we could not readily identify and validate a TAM-specific antibody to allow such measurements.

> ***Reviewer #3 (Recommendations For The Authors):***
>
> *(1) A clear description of the GEM model experiments will be helpful in interpreting the data as it is unclear what age the PTEN or MYC mice were when therapy was started. PTEN are generally intrinsically resistant to ARi whereas MYC are robustly sensitive.*
>
> *(2) Prostate organoid technology of the GEM prostate cell, and normal prostate cells may allow for a better evaluation of which basal stem-like cells are expressing TNF - dissecting out normal basal from cancer with basal-like properties.*
>
> *(3) Experiments to demonstrate targeting inhibition should be performed for AR and TNF inhibition. Especially across the spectrum of TNF blockade timing given the differences in proposed responsiveness over an acute change in dosing schedule.*
>
> *(4) Detailed histology and pathologic evaluation should be provided to characterize the impact on cancer and TME as well as normal prostate mixed in these tumors.*
>
> *(5) Prostate organoid development with genetic manipulation (PTEN ko) and transplant back into immunocompetent mice may provide experiments to prove causality and*

> *address the impact on the immune microenvironment.*
>
> *(6) The descriptive of regression and recurrence need to be defined as based on the kinetics and presented data this seems to be associated with minimal responsiveness and progression from a substantial volume of persistent cells.*
>
> *(7) The authors should also explore the impact of TNF inhibition on the cancer cell directly and evaluate downstream PI3K signaling.*

Responding to this set of recommendations: A number of these recommendations (R3-7, -9, -12) are similar or identical to those already noted in Reviewer 3's public review and have been addressed above. The remaining recommendations (R3-8, -10, -11; organoids, histological approaches to the TME, etc.) are potentially interesting experimental approaches but beyond the scope of the current manuscript.

> ***Reviewer #4 (Recommendations For The Authors):***
>
> *Major comments:*
>
> *(1) Figure 1A-B: While the decrease in tumor growth post-castration is apparent, the increase in tumor growth that has been designated as the point of androgen-independence is a mild increase from the 28 measurements and would benefit from statistical support. Further time points demonstrating that the tumors continue to increase in size would better support the claim that these tumors appropriately model disease recurrence.*

This data meets our criteria for recurrence (outlined in the M&M and in the legend to Table S2).

> *(2) Figure 2A: Statistical analysis should be performed and why is this figure shown twice (also in the S2A right panel)?*

We added statistical analysis to the legend of Fig. 2A. The data from Fig 2 (C4-2 cell line) is replicated in Supplementary Fig S2 to allow the reader to directly compare the response of the C4-2 cell line with the response of the LNCaP cell line.

> *(3) Figure 4A: Non-castrate + etan control is needed here. Also, the data should be statistically assessed.*

Regarding non-castrate controls, see our response to R4-2. Statistical analysis has been added - see Supplementary Table S3.

> *(4) It appears that at least two of the mice shown in Figure 5C have the same level of disease recurrence as was demonstrated in Figure 1B, yet the analysis defines recurrence in 0/6 mice.*

Again, similar to R4-7, None of the mice in Figure 5C meet our criteria for recurrence (outlined in the M&M and in the legend to Table S2).

> *(5) The text for Figure 5D states that vehicle-treated tumors (red) regress then recur while mice pre-treated with a CCR2 antagonist (blue) don't recur, but in the figure, these groups appear to be reversed. In addition, it would be good to have noncastrate + CCR2a control for Figure 5C and 5D.*

We corrected the labeling error in the legend to Figure 5.

> *(6) It would be good to validate major RNAseq findings using orthogonal approaches.*

We agree that it is valuable to validate our findings but these experiments are beyond the scope of the manuscript

> *(7) Figure 7B is quite puzzling. It appears to show the opposite of what was written.*

We thank the reviewer for bringing this error to our attention. Our internal review of previous versions of the manuscript showed that the corresponding author (JJK) inadvertently mis-edited this figure when preparing the BioRxiv submission. Figure 7B has been corrected and now aligns with the Results text. We have also appended a PDF documenting the editing error/ mistake.

> *(8) Figure 8: This experiment appears to have been done without replicates making the current interpretation questionable.*

A more detailed scRNAseq analysis of the GEMM response to castration (with replicated) is already underway. The analysis in Fig. 8 includes 1000's of cells, capturing the variation in mRNA levels. However, it does not capture animal-to-animal variation. Given the supporting role of this data in this manuscript, we believe that the single animal approach is adequate in this case.

> *(9) The level of detail included in the mechanism described in Figure S8 is not supported by the work shown.*

Fig. S8 is not presented as a summary of our findings but as a model that is consistent with our data - since it is by definition somewhat speculative, we present it in the supplementary data.

> *Minor Comments:*
>
> *(1) Figure 6S title is written incorrectly.*

We thank the reviewer for noticing this - we have corrected this in the revised manuscript.

> *(2) Images shown in Figure S7C need scale bars.*

These images are at 40X magnification - this has been added to the legend.

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